

# **MA2220: Statistical Methods Project**

## **Music & Mood Analysis Using Spotify Data**

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# 1 Introduction

Music plays a significant role in influencing human emotions and behavior. With the rise of music streaming platforms such as Spotify, large-scale datasets containing detailed musical characteristics are now available for statistical analysis. These datasets make it possible to study how measurable audio features relate to the emotional tone of songs.

This project focuses on analyzing whether musical attributes such as tempo, energy, and danceability can explain variations in the emotional positivity of a song, represented by the variable *valence*.

## 2 Objective

The primary objective of this study is to examine whether musical attributes can explain variations in the emotional tone of songs using quantitative analysis.

## 3 Research Question

Can musical attributes such as tempo, energy, and danceability significantly explain variations in the emotional tone (valence) of a song?

## 4 Dataset Description

The dataset used in this study is the Spotify Tracks Dataset obtained from Kaggle. It contains approximately 114,000 songs and 21 variables. Each row represents a song while each column represents a specific audio feature or metadata attribute.

The values in the dataset are generated using Spotify's proprietary audio analysis system, which applies digital signal processing and machine learning techniques to raw audio signals. Features such as tempo are detected using beat tracking algorithms, while attributes such as energy, danceability, and valence are estimated using machine learning models trained on musical characteristics.

### 4.1 Key Variables

- **Tempo:** Speed of the song measured in Beats Per Minute (BPM)
- **Energy:** Intensity and activity level of the song
- **Danceability:** Suitability of a song for dancing
- **Valence:** Emotional positivity of the song

## 5 Data Preprocessing

The dataset was cleaned and prepared before analysis using the following steps:

- Removal of missing values
- Selection of relevant variables
- Verification of numeric data types

The following variables were selected for analysis:

```
1 music <- read.csv("music.csv")
2 music <- na.omit(music)
3 colSums(is.na(music))
4 data <- music[, c("tempo", "energy", "danceability", "valence")]
```

## 6 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the distributions of variables and relationships between them.

### 6.1 Summary Statistics

Listing 1: Summary Statistics of Selected Variables

```
1 summary(data)
```

#### Output: Summary Statistics

|          | tempo   | valence        | energy         | danceability   |
|----------|---------|----------------|----------------|----------------|
| Min.     | : 0.00  | Min. :0.0000   | Min. :0.0000   | Min. :0.0000   |
| 1st Qu.: | 99.22   | 1st Qu.:0.2600 | 1st Qu.:0.4720 | 1st Qu.:0.4560 |
| Median   | :122.02 | Median :0.4640 | Median :0.6850 | Median :0.5800 |
| Mean     | :122.15 | Mean :0.4741   | Mean :0.6414   | Mean :0.5668   |
| 3rd Qu.: | 140.07  | 3rd Qu.:0.6830 | 3rd Qu.:0.8540 | 3rd Qu.:0.6950 |
| Max.     | :243.37 | Max. :0.9950   | Max. :1.0000   | Max. :0.9850   |

The summary statistics provide information regarding the range, mean, median, and quartiles of each variable. The dataset shows reasonable distributions with no extreme irregularities.

## 6.2 Histogram of Danceability

Listing 2: R Code for Danceability Histogram

```
1 hist(music$danceability,  
2     breaks = 30,  
3     col = "#55A868",  
4     border = "white",  
5     main = "Danceability",  
6     xlab = "Danceability",  
7     ylab = "Count")
```

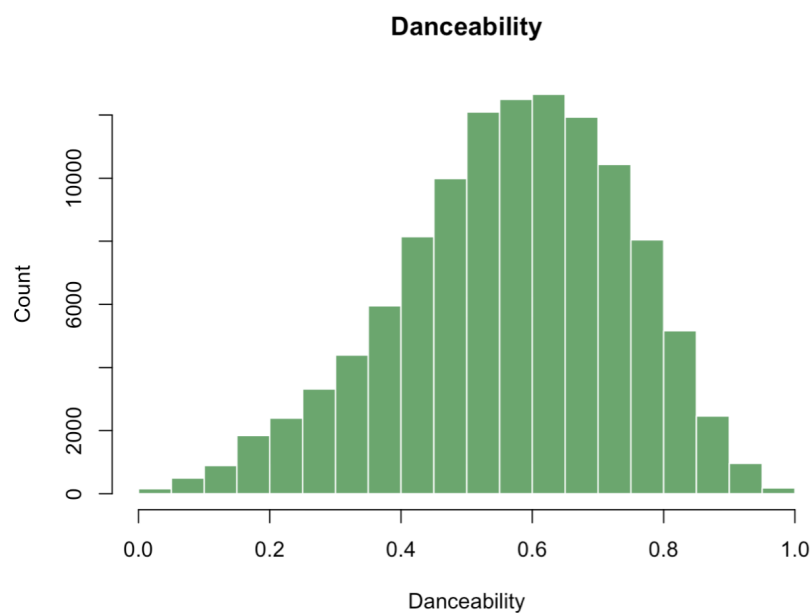


Figure 1: Histogram of Danceability

### Observations:

- The distribution of danceability is approximately bell-shaped with a slight left skew, indicating that most tracks in the dataset are moderately to highly danceable.
- The majority of songs fall between 0.40 and 0.80, suggesting that rhythmically engaging music dominates the dataset.
- Extremely low and extremely high danceability values are relatively rare, indicating limited representation of highly unconventional or highly structured tracks.
- The wider spread among lower danceability values suggests that less danceable songs vary more in musical style and structure compared to highly danceable tracks.

## 6.3 Histogram of Energy

Listing 3: R Code for Energy Histogram

```
1 hist(music$energy,  
2     main = "Histogram of Energy",  
3     xlab = "Energy",  
4     col = "tomato",  
5     border = "white")
```

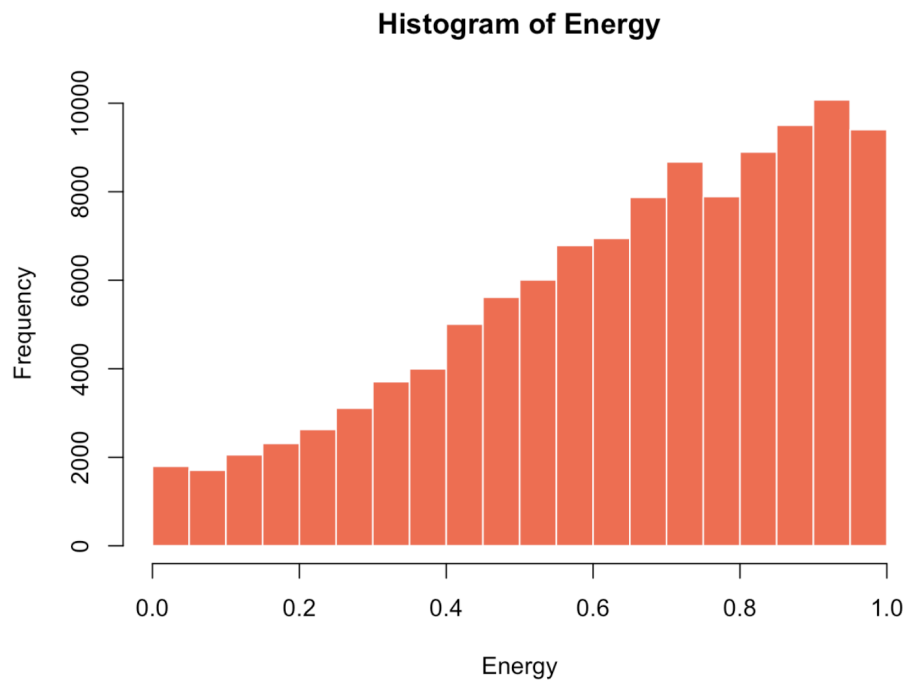


Figure 2: Histogram of Energy

### Observations:

- The energy distribution is slightly left-skewed, indicating that high-energy tracks dominate the dataset.
- Most songs have energy values between 0.60 and 1.0, reflecting the prevalence of energetic genres such as pop, EDM, hip-hop, and rock.
- Low-energy tracks are comparatively underrepresented, suggesting that calm or minimal acoustic music forms a smaller portion of the dataset.
- The gradual rise toward higher energy values rather than a symmetric bell-shaped curve indicates that energy is broadly distributed across mid-to-high intensity levels.

## 6.4 Histogram of Valence

Listing 4: R Code for Valence Histogram

```
1 hist(music$valence,  
2     breaks = 30,  
3     col = "#C44E52",  
4     border = "white",  
5     main = "Valence",  
6     xlab = "Valence",  
7     ylab = "Count")
```

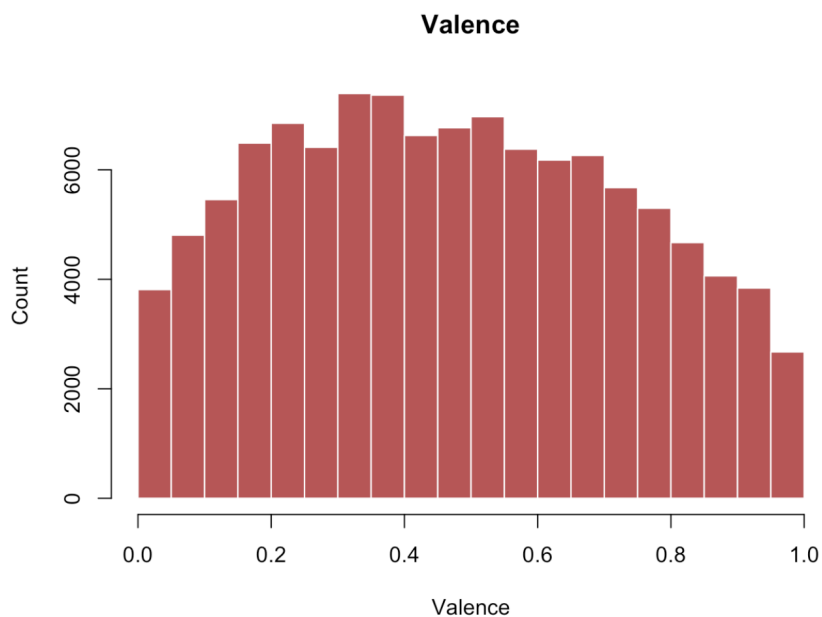


Figure 3: Histogram of Valence

### Observations:

- The distribution of valence is relatively broad and moderately bell-shaped, indicating that the dataset contains songs with a wide range of emotional tones.
- Mid-range valence values dominate the dataset, suggesting that emotionally neutral or mixed-emotion tracks are most common.
- Extremely positive and extremely negative songs are less frequent, indicating that emotional extremes are relatively uncommon within the dataset.
- Compared to other variables, valence appears more evenly distributed, making it a useful variable for statistical modelling and prediction.

## 6.5 Histogram of Tempo

Listing 5: R Code for Tempo Histogram

```
1 hist(music$tempo,  
2     main = "Histogram of Tempo",  
3     xlab = "Tempo (BPM)",  
4     col = "steelblue",  
5     border = "white")
```

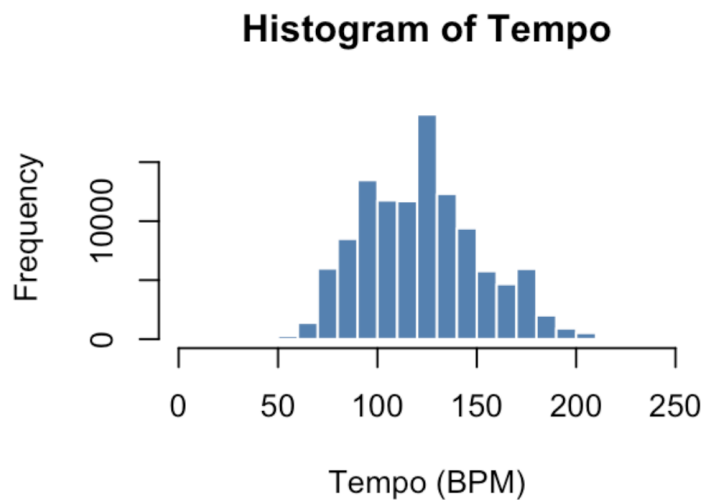


Figure 4: Histogram of Tempo

### Observations:

- The tempo distribution is slightly right-skewed, with the highest concentration of songs occurring between 120–130 BPM.
- Most tracks fall within the 80–160 BPM range, which aligns with the tempo range commonly observed in mainstream music genres.
- Extremely slow and extremely fast tracks are rare, indicating limited representation of niche musical styles.
- Although many songs cluster around similar tempo ranges, tempo alone may not strongly determine the emotional tone of a song.

## 6.6 Tempo vs Valence

Listing 6: R Code for Tempo vs Valence Scatter Plot

```
1 plot(music$tempo, music$valence,  
2     main = "Scatter Plot: Tempo vs Valence",  
3     xlab = "Tempo (BPM)",  
4     ylab = "Valence",  
5     col = "steelblue",  
6     pch = 16,  
7     cex = 0.5)
```

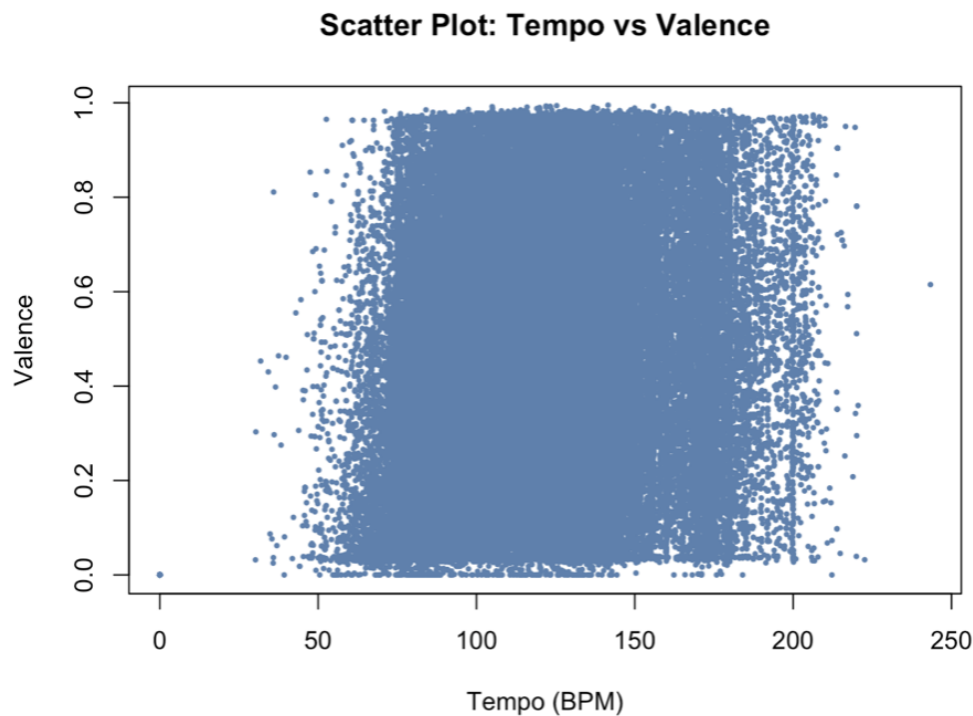


Figure 5: Scatter Plot of Tempo vs Valence

#### Observations:

- The scatter plot shows a very weak positive linear relationship between tempo and valence, supported by the low correlation coefficient of approximately 0.08.
- Songs across nearly all tempo ranges exhibit both high and low valence values, indicating that tempo alone is not a strong predictor of emotional tone.
- Most observations are concentrated between 60–200 BPM, reflecting the dominant tempo range within the dataset.
- The weak relationship suggests that other musical characteristics such as danceability, harmony, lyrics, and instrumentation may play a more important role in

determining mood.

## 6.7 Energy vs Valence

Listing 7: R Code for Energy vs Valence Scatter Plot

```
1 plot(music$energy, music$valence,  
2     main = "Energy vs Valence",  
3     xlab = "Energy",  
4     ylab = "Valence",  
5     col = rgb(0.2, 0.5, 0.8, alpha = 0.3),  
6     pch = 16,  
7     cex = 0.5)  
8 abline(lm(valence ~ energy, data = music),  
9       col = "red", lwd = 2)
```

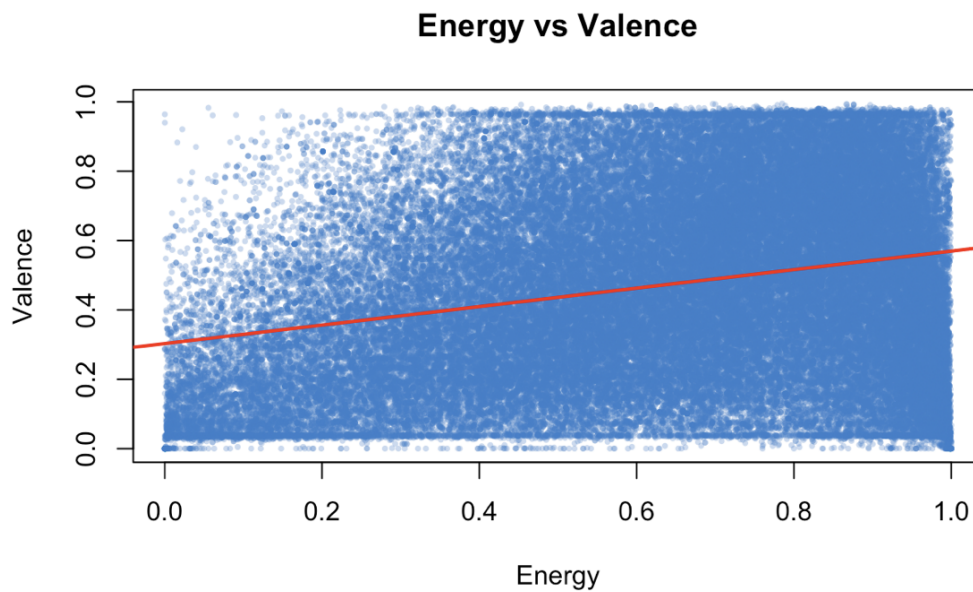


Figure 6: Scatter Plot of Energy vs Valence

### Observations:

- The scatter plot shows a moderate positive relationship between energy and valence, indicating that more energetic songs tend to exhibit higher emotional positivity.
- Songs with low energy values are more concentrated in lower valence ranges, while high-energy songs are more frequently associated with higher valence.

- Although the relationship is positive, considerable variation still exists, suggesting that energy alone does not completely determine mood.
- The distribution of points indicates that energetic songs are generally associated with more emotionally uplifting musical characteristics.

## 6.8 Danceability vs Valence

Listing 8: R Code for Danceability vs Valence Scatter Plot

```

1 plot(music$danceability, music$valence,
2       main = "Danceability vs Valence",
3       xlab = "Danceability",
4       ylab = "Valence",
5       col = rgb(0.2, 0.5, 0.8, alpha = 0.3),
6       pch = 16,
7       cex = 0.5)
8 abline(lm(valence ~ danceability, data = music),
9         col = "red", lwd = 2)

```

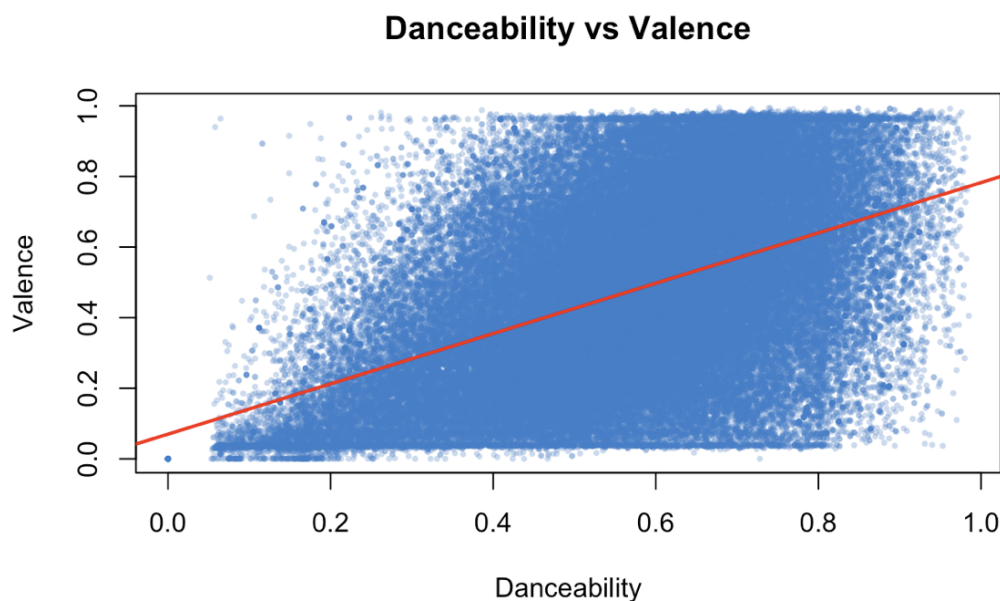


Figure 7: Scatter Plot of Danceability vs Valence

### Observations:

- The scatter plot demonstrates a moderate positive relationship between danceability and valence, supported by the relatively high correlation coefficient observed in

the correlation matrix.

- Songs with higher danceability values tend to exhibit higher valence, suggesting that rhythmically engaging tracks are generally perceived as more emotionally positive.
- Lower danceability songs show greater variation in valence, whereas highly danceable songs are more consistently associated with positive mood.
- Among the selected musical features, danceability appears to have the strongest visual relationship with valence and is likely to contribute significantly in regression analysis.

## 6.9 Correlation Matrix

Listing 9: R Code for Correlation Matrix Visualization

```
1 # Create Correlation Matrix
2 cor_matrix <- cor(data[, c("tempo",
3                             "energy",
4                             "danceability",
5                             "valence")],
6                  use = "complete.obs")
7
8 # Correlation Plot
9 corrplot(cor_matrix,
10          method = "color",
11          type = "upper",
12          addCoef.col = "black",
13          number.cex = 0.8,
14          tl.col = "black",
15          tl.srt = 20,
16          col = colorRampPalette(c("#4C72B0",
17                                   "white",
18                                   "#C44E52"))(200),
19          title = "Correlation Matrix of Music Variables",
20          mar = c(0,0,2,0))
```

### Correlation Matrix of Music Variables

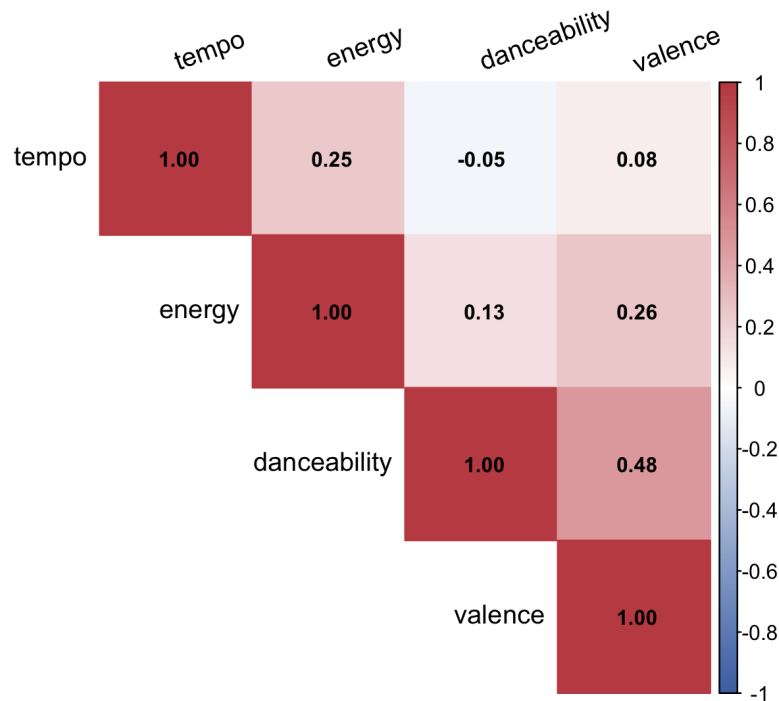


Figure 8: Correlation Matrix

#### Observations:

- Danceability and valence exhibit the strongest positive relationship ( $r = 0.48$ ), suggesting that more danceable songs tend to sound more emotionally positive.
- Energy and valence show a moderate positive correlation ( $r = 0.26$ ), indicating that energetic songs are generally associated with more positive emotional characteristics.
- Tempo and energy display a weak positive correlation ( $r = 0.25$ ), suggesting that faster songs may feel slightly more intense, although the relationship is not strong.
- Tempo and danceability are nearly uncorrelated ( $r = -0.05$ ), demonstrating that danceability depends more on rhythmic structure and groove than on speed alone.
- Overall, danceability appears to be the strongest predictor of valence among the selected variables and is likely to contribute significantly in regression analysis.

## 6.10 Side-by-Side Boxplots of Music Variables

Tempo values were standardized before visualization because tempo is measured on a much larger numerical scale compared to the other variables, which range between 0 and 1.

Listing 10: R Code for Side-by-Side Boxplots of Music Variables

```
1 # Scale Tempo
2 data$tempo_scaled <- as.numeric(scale(data$tempo))
3
4 # Side-by-Side Boxplots
5 boxplot(data[, c("danceability",
6                 "energy",
7                 "valence",
8                 "tempo_scaled")],
9         main = "Side-by-Side Boxplots of Music Variables",
10        ylab = "Scaled Values",
11        col = c("lightblue",
12               "lightpink",
13               "lightgreen",
14               "gold"),
15        border = "black")
```

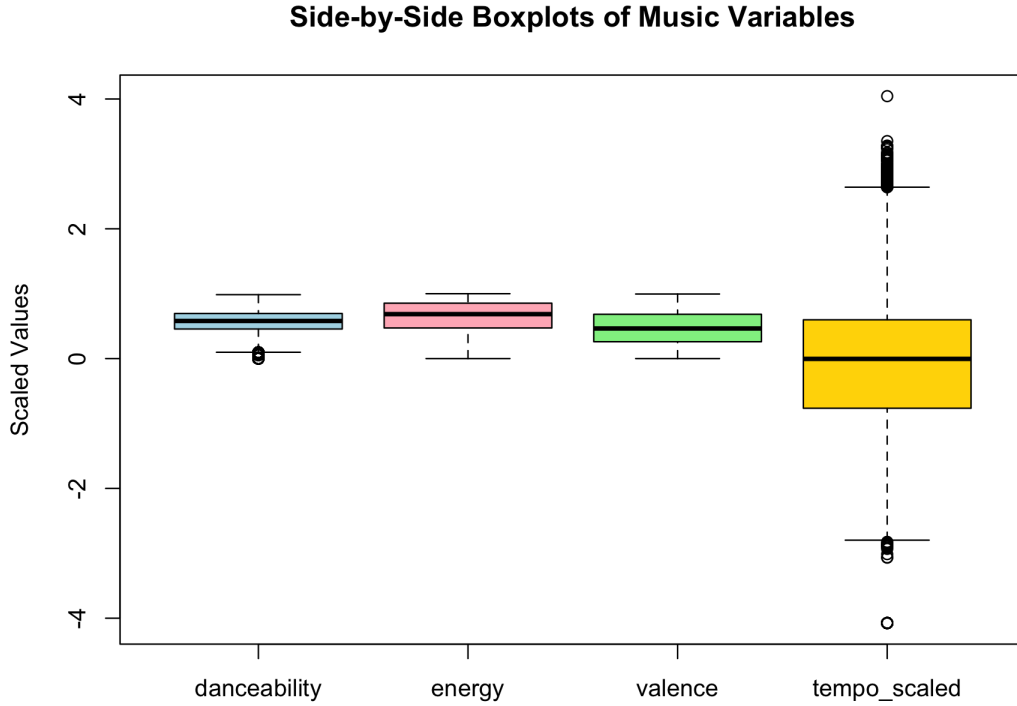


Figure 9: Side-by-Side Boxplots of Music Variables

### Observations

- The boxplots indicate noticeable variability across the selected musical variables.
- Tempo exhibits the largest spread and several visible outliers, indicating substantial variation in song speed within the dataset.
- Energy shows a relatively concentrated distribution toward higher values, supporting earlier observations from the histogram analysis.
- Danceability and valence display comparatively balanced distributions with moderate variability.

## 7 Methodology Overview

### 7.1 Statistical Analysis of Tempo and Valence

An independent two-sample t-test was conducted to compare the mean valence between slow-tempo and fast-tempo songs. Correlation analysis was additionally performed to measure the strength and direction of the linear relationship between tempo and valence.

Since certain variables, particularly tempo and energy, exhibited skewed distributions during exploratory data analysis, a Mann–Whitney U Test was also performed as a non-parametric alternative to validate the robustness of the results.

## 7.2 Multiple Linear Regression

Multiple Linear Regression was used to examine the effect of tempo, energy, and danceability on valence and to identify which musical features contribute most significantly to predicting emotional tone.

# 8 Hypothesis Testing: Association Between Tempo and Song Mood

This section examines whether tempo has a statistically significant relationship with song valence using parametric and non-parametric statistical tests.

## 8.1 Hypotheses

### Null Hypothesis ( $H_0$ ):

There is no significant association between tempo and valence of songs.

### Alternative Hypothesis ( $H_1$ ):

There is a significant association between tempo and valence in songs.

## 8.2 Methodology

To further examine the association between tempo and valence, songs were divided into two groups using the median tempo.

- **Slow Songs:** Songs with tempo lower than the median tempo
- **Fast Songs:** Songs with tempo greater than or equal to the median tempo

The median tempo of the dataset was found to be:

122.02

An independent (Welch) two-sample t-test was performed to compare the average valence between the two tempo groups.

In addition, correlation analysis was conducted to measure the strength and direction of the relationship between tempo and valence.

## 8.3 Independent Two-Sample t-test

### 8.3.1 R Code

Listing 11: R Code for Independent Two-Sample t-test

```
1 # Create Tempo Groups
2 median_tempo <- median(data$tempo)
3
4 data$tempo_group <- ifelse(data$tempo < median_tempo,
5                             "Slow",
6                             "Fast")
7
8 data$tempo_group <- as.factor(data$tempo_group)
9
10 # Perform t-test
11 test_result <- t.test(valence ~ tempo_group, data = data)
12 print(test_result)
13
14 # Calculate Mean Valence for Each Tempo Group
15 aggregate(valence ~ tempo_group,
16            data = data,
17            mean)
```

### 8.3.2 Output

Welch Two Sample t-test

```
data:  valence by tempo_group
t = 10.225, df = 113998, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Fast and
group Slow is not equal to 0
95 percent confidence interval:
 0.01268676 0.01870404
sample estimates:
mean in group Fast    mean in group Slow
    0.4819148         0.4662194
```

## 8.4 Interpretation

- The dataset was almost evenly divided into two tempo categories, with 57,008 fast songs and 56,992 slow songs.

- Songs classified under the fast-tempo group recorded an average valence of 0.482.
- Slow-tempo songs showed a comparatively lower average valence of 0.466.
- The independent two-sample t-test produced a t-statistic of 10.225.
- The degrees of freedom for the test were 113998.
- The p-value obtained from the test was less than  $2.2 \times 10^{-16}$ .

The independent two-sample t-test produced a p-value significantly lower than the significance level of 0.05. Therefore, the null hypothesis was rejected.

This suggests a statistically significant association between tempo and valence.

The higher average valence observed among fast-tempo songs suggests that songs with greater speed are generally associated with more positive emotional characteristics.

However, the difference between the two groups remains relatively small, indicating a weak practical effect despite strong statistical significance and suggesting that tempo alone is not a strong determinant of musical mood. Given the very large dataset size, even small differences can become statistically significant.

## 8.5 Correlation Analysis

Correlation analysis was performed to measure the strength and direction of the linear relationship between tempo and valence. The correlation coefficient helps determine whether changes in song tempo are associated with changes in emotional positivity.

### 8.5.1 R Code

Listing 12: R Code for Correlation Analysis

```
1 # Perform Correlation Analysis
2 correlation <- cor(data$tempo, data$valence)
3 print(correlation)
```

### 8.5.2 Output

```
0.07827307
```

### 8.5.3 Interpretation

The correlation coefficient between tempo and valence was found to be approximately 0.0783.

This indicates a weak positive linear association between the two variables, suggesting that songs with higher tempo tend to exhibit slightly higher valence.

However, the correlation strength is relatively low, implying that tempo alone is not a strong predictor of emotional tone.

While the large dataset of 114,000 songs makes even small differences statistically significant, the weak overall relationship indicates that other musical characteristics—such as danceability, energy, lyrics, and instrumentation—likely play a more substantial role in determining song mood.

## 8.6 Mann–Whitney Test

To further validate the robustness of the hypothesis testing results, a Mann–Whitney U Test (Wilcoxon Rank Sum Test) was also performed.

Unlike the independent t-test, the Mann–Whitney test is a non-parametric statistical test that does not assume normal distribution of data. This makes it suitable for datasets that may exhibit skewness or non-normal behavior.

### 8.6.1 R Code

Listing 13: R Code for Mann–Whitney U Test

```
1 # Mann-Whitney Test
2 wilcox_result <- wilcox.test(valence ~ tempo_group,
3                               data = data)
4
5 print(wilcox_result)
```

### 8.6.2 Output

```
Wilcoxon rank sum test with continuity correction
```

```
data: valence by tempo_group
```

```
W = 1.677e+09, p-value < 2.2e-16
```

```
alternative hypothesis: true location shift is not equal to 0
```

### 8.6.3 Interpretation

The test produced the following result:

$$W = 1.677 \times 10^9$$

with a p-value less than:

$$2.2 \times 10^{-16}$$

Since the p-value is significantly lower than 0.05, the null hypothesis was again rejected.

This confirms that difference in valence between the two tempo groups remains statistically significant even under a non-parametric testing approach.

Therefore, the Mann–Whitney test supports the conclusions obtained from the t-test.

## 8.7 Box Plot Visualisation

### 8.7.1 R Code

Listing 14: R Code for Boxplot

```
1 # Create Boxplot
2 boxplot(valence ~ tempo_group,
3         data = data,
4         main = "Valence Distribution for Slow and Fast Songs",
5         xlab = "Tempo Group",
6         ylab = "Valence",
7         col = c("skyblue", "lightgreen"),
8         border = "black",
9         notch = TRUE)
```

### 8.7.2 Output

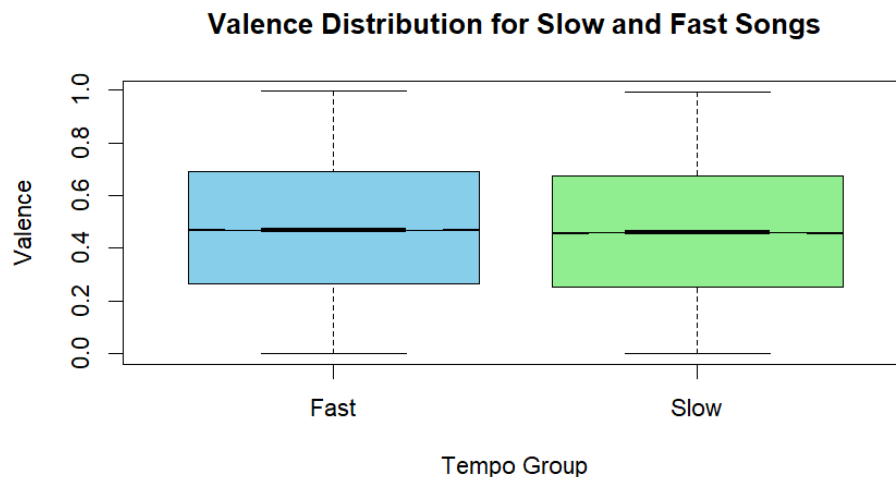


Figure 10: Valence Distribution for Slow and Fast Songs

### 8.7.3 Interpretation

The boxplot comparing slow and fast songs shows that the median valence of fast songs is slightly higher than that of slow songs.

Although the distributions overlap considerably, the fast-tempo group generally exhibits marginally higher valence values.

This supports the conclusion obtained from the t-test and correlation analysis that higher tempo songs tend to exhibit slightly more positive emotional characteristics.

The considerable overlap between the two distributions further indicates that the effect of tempo on valence is relatively weak.

## 8.8 Discussion

The positive association between tempo and valence suggests that songs with higher tempo are generally associated with slightly more positive emotional characteristics.

However, the weak correlation coefficient indicates that tempo alone is not a strong predictor of mood. Other musical features such as lyrics, genre, rhythm patterns, instrumentation, and harmony may also contribute significantly to the emotional perception of a song.

Therefore, while tempo has a measurable association with valence, it should not be considered the only factor affecting musical mood.

## 8.9 Limitations

- Valence may not perfectly capture human emotional perception of music.
- The analysis considered only a limited number of musical features.
- Statistical significance does not necessarily imply a strong practical effect, particularly in very large datasets.
- Other musical and cultural factors affecting mood were not included in this analysis.
- The division of songs into slow and fast tempo groups using the median tempo may oversimplify the continuous nature of tempo variation.

## 8.10 Conclusion

The analysis concluded that tempo has a statistically significant association with song valence.

Fast songs were observed to have slightly higher average valence than slow songs, indicating that higher tempo songs tend to exhibit slightly more positive emotional characteristics.

## 9 Regression Analysis

Regression analysis is a statistical technique used to investigate and model the relationship between a response variable and one or more predictor variables. In this project, we use Multiple Linear Regression to analyze the relationship between tempo, energy, and danceability with valence. The model aims to identify which audio features significantly contribute to predicting mood.

### 9.1 Regression Equation

We define the model as:

$$\text{Valence} = \beta_0 + \beta_1(\text{Tempo}) + \beta_2(\text{Energy}) + \beta_3(\text{Danceability}) + \epsilon$$

where,

- $\beta_0$  = intercept
- $\beta_1, \beta_2, \beta_3$  = regression coefficients
- $\epsilon$  = random error term

Before constructing the Multiple Linear Regression model, the following assumptions were assessed primarily using diagnostic plots and residual analysis

- **Linearity:** A linear relationship exists between the predictor variables and the response variable (valence).
- **Independence of Errors:** The error terms are assumed to be independent and uncorrelated with one another.
- **Homoscedasticity:** The variance of the error terms is assumed to remain constant across all levels of the predictor variables.
- **Normality of Residuals:** The residuals are assumed to follow a normal distribution.
- **Absence of Multicollinearity:** The predictor variables are assumed to not be highly correlated with each other.

### 9.2 R Code:

Listing 15: Construction of Multiple Linear Regression Model

```
1
2 # Select required variables
3 data <- data[, c("valence", "tempo", "energy", "danceability")]
4
5 # Build Multiple Linear Regression model
6 model <- lm(valence ~ tempo + energy + danceability, data = data)
7
8 # Display regression summary
9 summary(model)
```

Listing 16: Residual Plot

```
1 # Residual Plot
2 png("residual_plot.png", width = 900, height = 700)
3
4 plot(model$fitted.values,
5      model$residuals,
6      main = "Residual Plot",
7      xlab = "Fitted Values",
8      ylab = "Residuals",
9      col = "cyan3",
10     pch = 16)
11
12 abline(h = 0,
13        col = "darkcyan",
14        lwd = 2)
15
16 dev.off()
```

Listing 17: Histogram of Residuals

```
1 # Histogram of Residuals
2 png("histogram_residuals.png", width = 900, height = 700)
3
4 hist(model$residuals,
5      main = "Histogram of Residuals",
6      xlab = "Residuals",
7      col = "cyan3",
8      border = "darkcyan")
9
10 dev.off()
```

Listing 18: Normal Q-Q Plot

```

1 # Normal Q-Q Plot
2 png("qqplot.png", width = 900, height = 700)
3
4 qqnorm(model$residuals,
5       main = "Normal Q-Q Plot",
6       pch = 1)
7
8 qqline(model$residuals,
9       col = "cyan3",
10      lwd = 2)
11
12 dev.off()

```

Listing 19: Actual vs Predicted Values

```

1 # Actual vs Predicted Plot
2 png("actual_vs_predicted.png", width = 900, height = 700)
3
4 predicted <- predict(model)
5
6 plot(data$valence,
7      predicted,
8      main = "Actual vs Predicted Values",
9      xlab = "Actual Valence",
10     ylab = "Predicted Valence",
11     col = "cyan3",
12     pch = 16)
13
14 abline(0, 1,
15       col = "darkcyan",
16       lwd = 2)
17
18 dev.off()

```

### 9.3 Regression Output

The `summary(model)` function in R was used to obtain the regression coefficients, standard errors, t-values, p-values, and goodness-of-fit measures for the Multiple Linear Regression model.

**Fitted Regression Equation:**

$$\widehat{Valence} = -0.09199 + 0.000481(Tempo) + 0.1896(Energy) + 0.6804(Danceability)$$

```

Residuals:
    Min       1Q   Median       3Q      Max
-0.75709 -0.15593 -0.00608  0.16346  0.97386

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.199e-02  3.645e-03  -25.24  <2e-16 ***
tempo        4.813e-04  2.269e-05   21.22  <2e-16 ***
energy       1.896e-01  2.725e-03   69.58  <2e-16 ***
danceability  6.804e-01  3.831e-03  177.59  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2216 on 113996 degrees of freedom
Multiple R-squared:  0.2694,    Adjusted R-squared:  0.2694
F-statistic: 1.401e+04 on 3 and 113996 DF,  p-value: < 2.2e-16

```

Figure 11: Output of the Multiple Linear Regression Model

### 9.3.1 Interpretation of Coefficients

The regression coefficients represent the effect of each predictor variable on valence while keeping the other variables constant. The positive coefficient for tempo indicates that an increase in song tempo is associated with a slight increase in valence. Similarly, the positive coefficient for energy suggests that songs with higher intensity and activity levels tend to have higher emotional positivity. Danceability showed the largest positive coefficient, indicating that highly danceable songs are generally associated with higher valence values.

### 9.3.2 P-Values

The p-values for tempo, energy, and danceability were all less than 0.05, indicating that the predictor variables are statistically significant. This suggests that each of these musical features contributes meaningfully to predicting valence in the regression model.

## 9.4 Model Evaluation Metrics

### 9.4.1 R Squared Values

The Multiple R-squared value of 0.2694 indicates that approximately 26.94% of the variation in valence can be explained by tempo, energy, and danceability. This suggests that while the selected predictors have a statistically significant relationship with valence, substantial variation in valence remains unexplained by the selected variables.

**Adjusted R Squared Value** The Adjusted R-squared value of 0.2694 accounts for the number of predictor variables included in the model and confirms that the selected variables contribute meaningfully to explaining variations in valence without unnecessary complexity.

### 9.4.2 Residual Standard Error

The Residual Standard Error (RSE) of 0.2216 represents the average difference between the observed valence values and the values predicted by the regression model. A relatively lower RSE indicates that the model predictions are reasonably close to the actual values.

### 9.4.3 F-statistic

The F-statistic evaluates the overall significance of the regression model. The extremely small p-value associated with the F-statistic ( $< 2.2 \times 10^{-16}$ ) indicates that the regression model is statistically significant as a whole, meaning that at least one predictor variable has a significant relationship with valence.

## 9.5 Diagnostic Plots

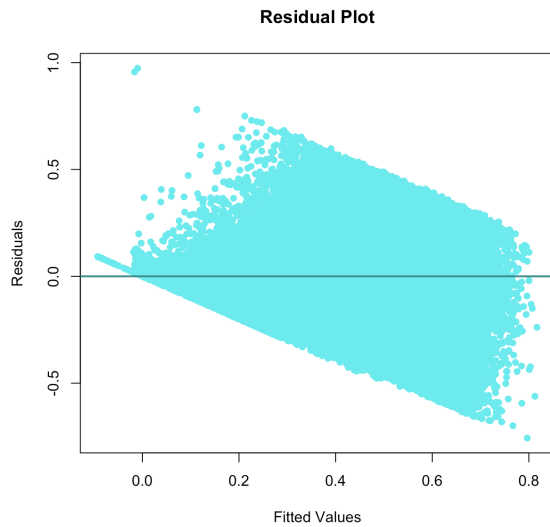


Figure 12: Residual Plot

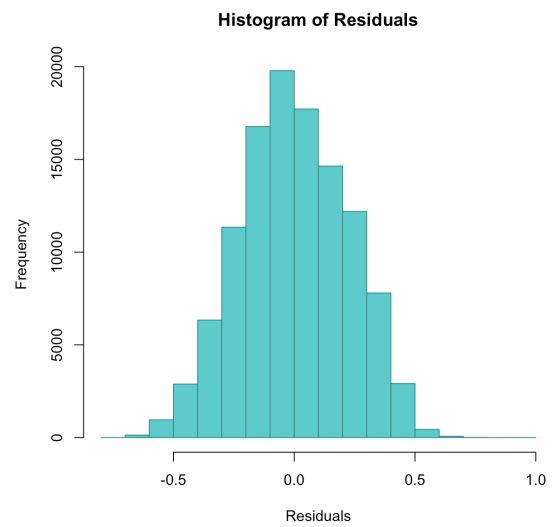


Figure 13: Histogram of Residuals

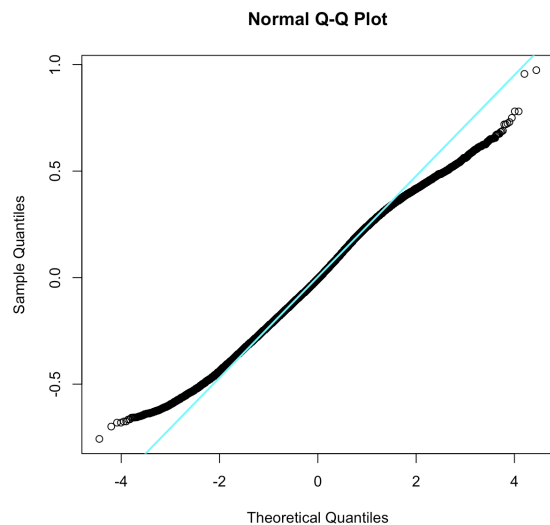


Figure 14: Normal Q-Q Plot

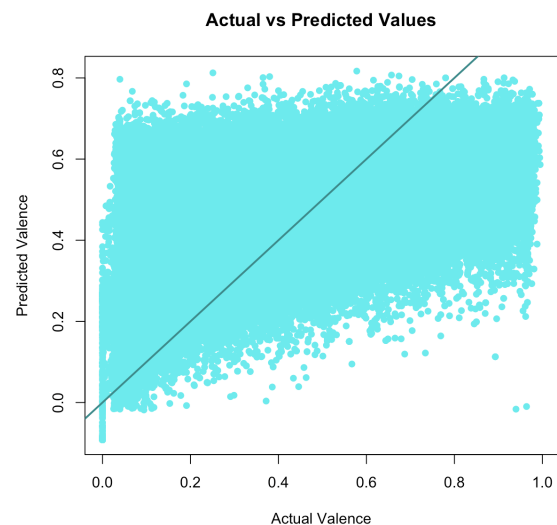


Figure 15: Actual vs Predicted Values

- **Residual Plot:** The residuals are distributed around zero although slight heteroscedasticity is visible
- **Histogram of Residuals:** The residuals appear approximately symmetric and centered around zero, indicating near-normal distribution.
- **Normal Q-Q Plot:** Most points lie close to the reference line, suggesting that residuals are approximately normal, with slight deviations in the tails.

- **Actual vs Predicted Plot:** The predicted values generally follow the trend of the observed valence values, suggesting that the model captures overall variation reasonably well.

## 9.6 Results

The Multiple Linear Regression model was used to examine the relationship between Spotify audio features and song valence. The regression results showed that tempo, energy, and danceability were all statistically significant predictors of valence, as indicated by their extremely small p-values ( $< 2.2 \times 10^{-16}$ ). Among the predictor variables, danceability showed the strongest positive influence on valence, followed by energy and tempo.

The model produced an  $R^2$  value of 0.2694, indicating that approximately 26.94% of the variation in valence can be explained by the selected audio features. The Residual Standard Error of 0.2216 suggests a reasonable fit between observed and predicted values. Additionally, the large F-statistic and its associated p-value confirmed that the regression model is statistically significant overall.

The diagnostic plots indicated that the residuals were approximately centered around zero and followed a near-normal distribution, suggesting that the assumptions of Multiple Linear Regression were reasonably satisfied. Overall, the analysis demonstrated that tempo, energy, and danceability contribute meaningfully to predicting the emotional tone of songs represented by valence.

## 10 Results and Discussion

The exploratory analysis indicates that danceability has the strongest association with valence among the selected variables. Songs that are more danceable tend to exhibit more positive emotional characteristics.

Energy also demonstrates a moderate positive association with valence, suggesting that energetic songs are generally associated with more positive moods.

Tempo shows a statistically significant but weak association with valence, indicating that song speed alone is not a strong determinant of emotional tone.

The hypothesis testing results showed that fast-tempo songs had slightly higher average valence compared to slow-tempo songs. However, the observed difference between the two groups was relatively small, indicating a weak practical effect despite strong statistical significance.

The correlation and regression analyses further suggest that danceability and energy contribute more strongly to variations in valence than tempo.

Overall, the findings indicate that measurable musical characteristics can provide meaningful insight into the emotional tone of songs, although a substantial portion of

variation in valence remains influenced by additional musical and contextual factors.

## 11 Conclusion

This study examined the association between musical audio features and emotional tone using Spotify data. The analysis confirmed that danceability and energy are positively associated with valence, while tempo shows a statistically significant but weaker positive association.

The hypothesis testing, correlation analysis, and multiple linear regression model consistently demonstrated that musical characteristics contribute meaningfully to variations in song mood. Among the selected variables, danceability emerged as the strongest predictor of valence, followed by energy and tempo.

The findings demonstrate that measurable audio features can effectively be used to analyse and predict the emotional tone of songs. Overall, the project highlights the usefulness of statistical methods and large-scale music datasets in understanding patterns in musical mood and listener perception.

## 12 Challenges

- Mood and emotional tone were represented using Spotify’s valence metric, which may not fully capture human emotional perception of music.
- Emotional interpretation of music is subjective
- The analysis focused only on selected audio features and did not include lyrical content, genre-specific influences, or cultural factors affecting musical mood.
- Interpreting statistical significance was challenging due to the very large sample size, where even small differences became statistically significant.

## 13 Future Scope

- Include additional audio features and metadata to perform more detailed analysis of musical mood.
- Apply advanced machine learning techniques to improve mood prediction and classification accuracy.
- Perform lyrical sentiment analysis to study how song lyrics contribute to emotional tone alongside audio characteristics.

- Develop music recommendation systems based on emotional tone and listener preferences.
- Use another dataset containing personality traits and combine it with music-related datasets to analyse how musical characteristics and listening preferences may relate to different personality types.

## 14 References

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- Sheldon Ross, Introduction to Probability and Statistics for Engineers and Scientists, Fifth Edition, Academic Press, 2014.
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- <https://www.codecademy.com/learn/learn-linear-regression-in-r/modules/linear-regression-in-r/cheatsheet>
- [https://guides.library.lincoln.ac.uk/mash/mann\\_whitney\\_R](https://guides.library.lincoln.ac.uk/mash/mann_whitney_R)

## A Complete R Code

```
1 install.packages("corrplot")
2 library(corrplot)
3
4 # Load Dataset
5 music <- read.csv("music.csv")
6
7 # Remove Missing Values
8 music <- na.omit(music)
9
10 # Select Required Variables
11 data <- music[, c("tempo",
12                  "energy",
13                  "danceability",
14                  "valence")]
15
16 # Check Missing Values
17 colSums(is.na(data))
18
19 #Summary Statistics
20 summary(data)
21
22
23 # Danceability Histogram
24 hist(music$danceability,
25       breaks = 30,
26       col = "#55A868",
27       border = "white",
28       main = "Danceability",
29       xlab = "Danceability",
30       ylab = "Count")
31
32
33 # Energy Histogram
34 hist(music$energy,
35       breaks = 30,
36       col = "tomato",
37       border = "white",
38       main = "Energy",
39       xlab = "Energy",
```

```

40     ylab = "Count")
41
42
43 # Valence Histogram
44 hist(music$valence,
45       breaks = 30,
46       col = "#C44E52",
47       border = "white",
48       main = "Valence",
49       xlab = "Valence",
50       ylab = "Count")
51
52
53 # Tempo Histogram
54 hist(music$tempo,
55       breaks = 30,
56       col = "steelblue",
57       border = "white",
58       main = "Tempo",
59       xlab = "Tempo (BPM)",
60       ylab = "Count")
61
62
63 # Tempo vs Valence Scatter Plot
64 plot(data$tempo,
65       data$valence,
66       main = "Scatter Plot: Tempo vs Valence",
67       xlab = "Tempo (BPM)",
68       ylab = "Valence",
69       col = "steelblue",
70       pch = 16,
71       cex = 0.5)
72
73
74 # Energy vs Valence Scatter Plot
75 plot(data$energy,
76       data$valence,
77       main = "Scatter Plot: Energy vs Valence",
78       xlab = "Energy",
79       ylab = "Valence",
80       col = "tomato",

```

```

81     pch = 16,
82     cex = 0.5)
83
84 # Danceability vs Valence Scatter Plot
85 plot(data$danceability,
86      data$valence,
87      main = "Scatter Plot: Danceability vs Valence",
88      xlab = "Danceability",
89      ylab = "Valence",
90      col = "#55A868",
91      pch = 16,
92      cex = 0.5)
93
94 # Create Correlation Matrix
95 cor_matrix <- cor(data[, c("tempo",
96                          "energy",
97                          "danceability",
98                          "valence")],
99                use = "complete.obs")
100
101 # Correlation Plot
102 corrplot(cor_matrix,
103          method = "color",
104          type = "upper",
105          addCoef.col = "black",
106          number.cex = 0.8,
107          tl.col = "black",
108          tl.srt = 20,
109          col = colorRampPalette(c("#4C72B0",
110                                "white",
111                                "#C44E52"))(200),
112          title = "Correlation Matrix of Music Variables",
113          mar = c(0,0,2,0))
114
115
116 # Scale Tempo
117 data$tempo_scaled <- as.numeric(scale(data$tempo))
118
119 # Side-by-Side Boxplots
120 boxplot(data[, c("danceability",
121                  "energy",

```

```

122         "valence",
123         "tempo_scaled"]],
124     main = "Side-by-Side Boxplots of Music Variables",
125     ylab = "Scaled Values",
126     col = c("lightblue",
127             "lightpink",
128             "lightgreen",
129             "gold"),
130     border = "black")
131
132 # Create Tempo Groups
133 median_tempo <- median(data$tempo)
134
135 data$tempo_group <- ifelse(data$tempo < median_tempo,
136                            "Slow",
137                            "Fast")
138
139 data$tempo_group <- as.factor(data$tempo_group)
140
141 #Independent Two-Sample t-test
142 test_result <- t.test(valence ~ tempo_group,
143                       data = data)
144 print(test_result)
145
146
147 # Mean Valence for Each Group
148 aggregate(valence ~ tempo_group,
149           data = data,
150           mean)
151
152 #Correlation Analysis
153 correlation <- cor(data$tempo,
154                   data$valence)
155 print(correlation)
156
157 #Mann-Whitney U Test
158 wilcox_result <- wilcox.test(valence ~ tempo_group,
159                              data = data)
160
161 print(wilcox_result)
162

```

```

163 #Box Plot for Hypothesis Testing
164 boxplot(valence ~ tempo_group,
165         data = data,
166         main = "Valence Distribution for Slow and Fast Songs",
167         xlab = "Tempo Group",
168         ylab = "Valence",
169         col = c("skyblue", "lightgreen"),
170         border = "black",
171         notch = TRUE)

```

Listing 20: Construction of Multiple Linear Regression Model

```

1
2 # Select required variables
3 data <- data[, c("valence", "tempo", "energy", "danceability")]
4
5 # Build Multiple Linear Regression model
6 model <- lm(valence ~ tempo + energy + danceability, data = data)
7
8 # Display regression summary
9 summary(model)

```

Listing 21: Residual Plot

```

1 # Residual Plot
2 png("residual_plot.png", width = 900, height = 700)
3
4 plot(model$fitted.values,
5      model$residuals,
6      main = "Residual Plot",
7      xlab = "Fitted Values",
8      ylab = "Residuals",
9      col = "cyan3",
10     pch = 16)
11
12 abline(h = 0,
13       col = "darkcyan",
14       lwd = 2)
15
16 dev.off()

```

Listing 22: Histogram of Residuals

```

1 # Histogram of Residuals
2 png("histogram_residuals.png", width = 900, height = 700)
3
4 hist(model$residuals,
5       main = "Histogram of Residuals",
6       xlab = "Residuals",
7       col = "cyan3",
8       border = "darkcyan")
9
10 dev.off()

```

Listing 23: Normal Q-Q Plot

```

1 # Normal Q-Q Plot
2 png("qqplot.png", width = 900, height = 700)
3
4 qqnorm(model$residuals,
5        main = "Normal Q-Q Plot",
6        pch = 1)
7
8 qqline(model$residuals,
9        col = "cyan3",
10        lwd = 2)
11
12 dev.off()

```

Listing 24: Actual vs Predicted Values

```

1 # Actual vs Predicted Plot
2 png("actual_vs_predicted.png", width = 900, height = 700)
3
4 predicted <- predict(model)
5
6 plot(data$valence,
7      predicted,
8      main = "Actual vs Predicted Values",
9      xlab = "Actual Valence",
10     ylab = "Predicted Valence",
11     col = "cyan3",
12     pch = 16)
13
14 abline(0, 1,
15        col = "darkcyan",

```

```
16     lwd = 2)
17
18 dev.off()
```